

Comments

In C, **comments** are used to write notes inside the program. They are **ignored by the compiler** and help make code easier to understand.

Types of Comments in C

1. Single-line Comment

- Starts with //
- Used for short explanations

```
#include <stdio.h>
int main() {
    // This is a single-line comment
    printf("Hello World");
    return 0;
}
```

Multi-line Comment

- Starts with /* and ends with */
- Used for longer explanations

```
#include <stdio.h>
int main() {
    /* This is a
       multi-line comment */
    printf("Hello Ashok");
    return 0;
}
```

Indentation

Indentation means adding **spaces or tabs at the beginning of lines** to clearly show the structure of the program.

It does **not affect compilation or output**, but it **improves readability**

Why Indentation is Important?

- Makes code easy to read
- Helps understand blocks (if, for, while)
- Reduces logical errors
- Improves debugging

Error Messages

In C programming, error messages indicate problems in your code. They are generally classified into 4 main types:

Syntax Errors (Compile-Time Errors)

Errors that violate the grammar rules of C. Detected by the compiler before execution

```
#include <stdio.h>
int main() {
    printf("Hello World); // missing closing quote
    return 0;
}
```

Semantic Errors (Compile-Time Errors)

Errors related to **meaning**, even if syntax is correct.

```
int x = "Hello"; // assigning string to int
```

Linking Error

A **linking error** occurs **after compilation**, during the **linking stage**, when the linker is unable to combine object files and libraries into a final executable code.

Suppose we declare `add()` but never provide its definition.

```
//Missing Function Definition
int add(int, int);
int main() {
    int result = add(10, 20);
    return 0;
}
```

Runtime Errors

Errors that occur **while the program is running**

```
#include <stdio.h>
int main() {
    int a = 10, b = 0;
    printf("%d", a / b); // division by zero
    return 0;
}
```

Logical Errors

Program runs **without error**, but gives **wrong output**.

```
#include <stdio.h>
int main() {
    int a = 5, b = 10;
    printf("%d", a - b); // wrong logic (wanted addition)
    return 0;
}
```